

AGENTiX systems

VPS Architecture & System Solution Design Document

A Single Source of Truth for Servers, Databases, Networking, and Security Protocols

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VPS Architecture & System Solution Design Document

This document serves as the **Single Source of Truth** for the production server infrastructure, databases, networking, and security configurations hosted on your VPS.



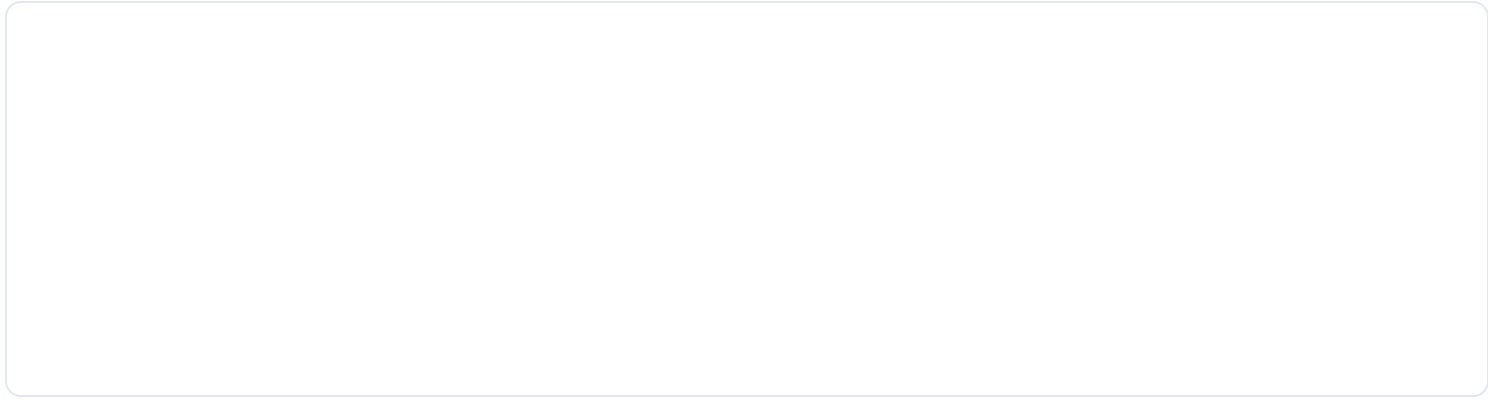
System Overview

- **Host Public IP:** 76.13.250.173
 - **Host OS:** Ubuntu Linux
 - **Management Layer:** Coolify v4 (Beta 463)
 - **SSL Edge Proxy:** Traefik v3.6 (coolify-proxy)
 - **Host Web Server:** Nginx (listening locally on port 8090 for ai-agentix.com)
 - **Process Manager:** PM2 (running ai-agentix-backend on Node.js v20)
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Network & Routing Architecture

All external traffic enters through Traefik (coolify-proxy), which handles Let's Encrypt SSL/TLS termination and routes traffic to internal Docker networks or host-level loopback interfaces.



Active Applications Registry

Application Name	Service Type	Deployment Path / Container	Internal Port	External Domain / Route
coolify-proxy	Traefik SSL Proxy	Docker: coolify-proxy	80 , 443	Direct edge routing
coolify	Management Panel	Docker: coolify	8000	Port 8000
nocodb	No-Code DB Layer	Docker: nocodb-swcsokowgkko8ogwcsw0c044	8080	nocodb-swcsokowgkko8ogwcs
pocketbase	No-Code Backend	Docker: pocketbase-jkc4oswoc00sw84wkgcsgk0c	8080	Managed under Coolify
pagarbook-bot	WhatsApp Bot API	Docker: pagarbook-bot-container	3000	pagarbook.aiagentixdev.com
kanban-crm	CRM Dashboard	Docker: kanban-crm	5000	crm.aiagentixdev.com
ai-agentix-frontend	React Static Frontend	Host Nginx: /var/www/ai-agentix/frontend/dist	8090	ai-agentix.com & ai-agentix.com
ai-agentix-backend	Node Express API	PM2: /var/www/ai-agentix/backend/server.js	3001	ai-agentix.com/api



Database Infrastructure

1. Central PostgreSQL Container

- **Container Name:** `w04cscwscsc880sc488cscg`
- **Docker DNS Alias:** `w04cscwscsc880sc488cscg` (resolves internally inside Docker `coolify` network)
- **Port:** `5432` (Exposed to host `0.0.0.0:5432`)
- **Superuser Username:** `postgres`
- **Superuser Password:** `AGENTiX@2025`
- **Active Databases & Mappings:**
 1. `client_pagarbook_vms_db` : Stores WhatsApp Bot data (templates, chat sessions, campaign logs, contacts). Connected to `pagarbook-bot-container` and **NocoDB**.
 2. `crm` : Stores Kanban CRM records (leads, pipelines, custom tags). Connected to `kanban-crm` and **NocoDB**.
 3. `postgres` : Central administration and system config database. Connected to **NocoDB**.

2. External Hostinger MySQL Database

- **Host:** `srv1988.hstgr.io`
- **Port:** `3306`
- **Username:** `u540387157_AI_Agentix`
- **Password:** `AGENTiX@2025`
- **Database Name:** `u540387157_AI_Agentix`
- **Active Tables (12):** `case_studies` , `awards` , `admins` , `clients` , `contacts` , `demo_bookings` , `resources` , `posts` , `subscribers` , `services` , `team_members` , `voice_agent_leads`
- **Connected Applications:**
 - `ai-agentix-backend` (running via PM2 on the VPS)
 - **NocoDB** (mapped as data source `ai_agentix_db`)

3. NocoDB Metadata Database (SQLite)

- **Path inside Container:** `/usr/app/data/noco.db`
- **Path on Host Volume:** `/var/lib/docker/volumes/swcsokowgkko8ogwcs0c044_nocodb-data/_data/noco.db`
- **User Sign-In:** `aiagentix2025@gmail.com`
- **User Password:** `AGENTiX@2025`
- **MCP Token:** `sf5oZ18JwSiwfiQZlALL52a7tYDtGfP7`
- **Configured Data Sources:**
 - `pagarbook_bot` (Postgres `client_pagarbook_vms_db`)
 - `kanban_crm` (Postgres `crm`)
 - `central_postgres` (Postgres `postgres`)
 - `ai_agentix_db` (Hostinger MySQL `u540387157_AI_Agentix`)

4. PocketBase Database (SQLite)

- **Path on Host Volume:** `/data/coolify/services/jkc4oswoc00sw84wkgcsgk0c/pb_data`
- **Dashboard FQDN:** Configured inside Coolify UI.



Security & Firewall Configurations

The host firewall uses `UFW` (Uncomplicated Firewall) with a strict default `DROP` policy. Only required ports are allowed:

1. UFW Rules List

- `22/tcp` (**OpenSSH**): Allowed from anywhere (admin shell access).
- `80/tcp` , `443/tcp` : Allowed from anywhere (public web traffic to Traefik).
- `5433/tcp` : Allowed from anywhere (management access).
- **Docker Network Loopback Rules:**

To allow the Traefik docker container (`10.0.1.11`) to forward requests to Nginx running on the host port `8090` , UFW explicitly permits TCP traffic from Docker subnets:

```
ufw allow proto tcp from 10.0.0.0/8 to any port 8090
ufw allow proto tcp from 172.16.0.0/12 to any port 8090
```

2. Nginx Host Configuration (`/etc/nginx/sites-available/ai-agentix`)

Nginx binds locally to port `8090` without SSL (as Traefik handles Let's Encrypt SSL at the edge). It matches requests to `ai-agentix.com` / `www.ai-agentix.com` :

- **Static Assets:** Serves React app files directly from `/var/www/ai-agentix/frontend/dist` .
- **API Proxying:** Proxies all requests to `/api/` to `http://localhost:3001/api/` .

3. Traefik Routing Configuration (`/data/coolify/proxy/dynamic/ai-agentix.yaml`)

Enforces HTTP-to-HTTPS redirection and proxies HTTPS traffic to Nginx on the host gateway IP `host.docker.internal` (`10.0.0.1`):

```
http:
  routers:
    ai-agentix-http:
      rule: "Host(`ai-agentix.com`) || Host(`www.ai-agentix.com`)"
      service: ai-agentix
      entryPoints:
        - http
      middlewares:
        - ai-agentix-redirect-to-https
    ai-agentix-https:
      rule: "Host(`ai-agentix.com`) || Host(`www.ai-agentix.com`)"
      service: ai-agentix
      entryPoints:
        - https
      tls:
        certResolver: letsencrypt
  middlewares:
    ai-agentix-redirect-to-https:
      redirectScheme:
        scheme: https
        permanent: true
  services:
    ai-agentix:
      loadBalancer:
        servers:
          - url: "http://host.docker.internal:8090"
```

Deprecatcd & Collapsed Resources (Cleaned Assets)

To optimize disk storage and memory consumption, the following legacy assets were fully decommissioned:

1. **nritrust** **Application:** Crashing PM2 process stopped and deleted. Source files at `/var/www/nritrust` removed.

2. **aisummitjaipur.aiagentixdev.com** Landing Page: Files at `/var/www/aisummitjaipur.aiagentixdev.com` and its host Nginx symlinks removed.
3. **Legacy Databases:** Dropped PostgreSQL databases `agentix` and `indiagrain_whatsapp_crm` (releasing space in the Postgres container).
4. **Outdated Backups:** Removed all historical dumps (`.dump` , `.sql` , `.tar.gz`) from `/root/` to recover host storage.
5. **Leftover Directories:** Deleted `/root/upload-api` and `/root/minio-data` .

Maintenance & Operations Playbook

1. Process Monitoring

To check on the state of the Node.js backend:

```
# View active processes and resource usages
pm2 list

# Watch logs in real time
pm2 logs ai-agentix-backend

# Restart backend process
pm2 restart ai-agentix-backend
```

2. Router Configurations (Adding new custom routing)

Traefik dynamic configuration files are located under `/data/coolify/proxy/dynamic/` . When adding or editing configurations (like `crm.yaml` or `ai-agentix.yaml`), Traefik will automatically hot-reload them. If a hard reload is required:

```
docker restart coolify-proxy
```

3. Nginx Operations

```
# Test configurations for syntax errors
nginx -t

# Reload configuration changes
systemctl reload nginx

# Restart service
systemctl restart nginx
```

4. Database Access (Manual PostgreSQL Connection)

```
# Enter the Postgres container command line
docker exec -it w04cscwscsc880sc488cscg psql -U postgres -d postgres
```